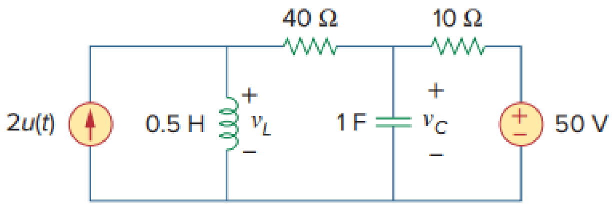
Chapter 8, Problem 31P

Problem

Consider the circuit in Fig. 8.79. Find *vL*(0+) and *vC*(0+).

Figure 8.79:



Step-by-step solution

Step 1 of 6

Refer to figure 8.79 in the textbook.

Define the input current.

$$2u(t)=\begin{cases}0\text{ A}, & t<0\\ 2\text{ A}, & t>0\end{cases}$$

Redraw the circuit for $t<0$ as shown in figure 1.

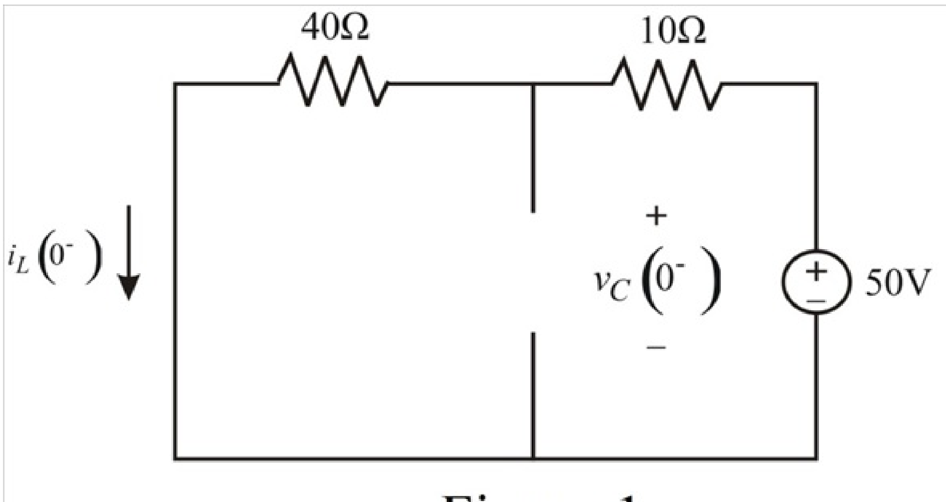


Figure 1

Step 2 of 6

Calculate the initial voltage across the capacitor.

Apply the voltage division rule to the circuit shown in figure 1.

$$\begin{aligned}v_C(0^-)&=\frac{40}{40+10}\times 50\\&=40\text{ V}\end{aligned}$$

The capacitor doesn't allow sudden changes in voltage.

$$v_C(0^-)=v_C(0^+)=40\text{ V}$$

Therefore, the voltage $v_C(0^+)$ is $\boxed{40\text{ V}}$.

Step 3 of 6

Calculate the initial current through the inductor.

Apply Ohm's law to the circuit shown in figure 1.

$$\begin{aligned} i_L\left(0^-\right) &= \frac{50}{40+10} \\ &= \frac{50}{50} \\ &= 1\text{ A} \end{aligned}$$

The inductor doesn't allow sudden changes in current.

$$i_L\left(0^-\right) = i_L\left(0^+\right) = 1\text{ A}$$

Step 4 of 6

Draw the circuit for $t = 0^+$ as shown in figure 2.

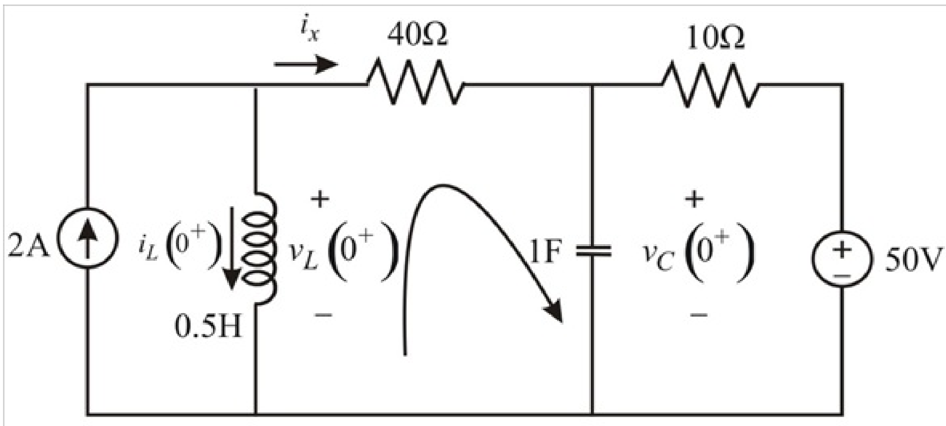


Figure 2

Step 5 of 6

Apply Kirchhoff's current law at the input node of the circuit shown in figure 2.

$$i_x + i_L\left(0^+\right) = 2$$

$$i_x = 2 - i_L\left(0^+\right)$$

Substitute 1 for $i_L\left(0^+\right)$.

$$\begin{aligned} i_x &= 2-1 \\ &= 1\text{ A} \end{aligned}$$

Step 6 of 6

Apply Kirchhoff's voltage law around the centre loop of the circuit shown in figure 2.

$$-v_L\left(0^+\right)+40i_x+v_C\left(0^+\right)=0$$

$$v_L\left(0^+\right)=40i_x+v_C\left(0^+\right)$$

Substitute 1 for i_x, and 40 for $v_C\left(0^+\right)$.

$$\begin{aligned} v_L\left(0^+\right) &= 40\times 1+40 \\ &= 80 \text{ V} \end{aligned}$$

Therefore, the value of $v_L\left(0^+\right)$ is $\boxed{80 \text{ V}}$.